Introductory molecular spectroscopy

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Introduction

This lecture is an introduction to molecular spectroscopy, with a focus on **optical spectroscopic techniques**. Non-optical techniques, such as electronic spectroscopy, γ-rays (Mössbauer spectroscopy), or magnetic nuclear resonance will not be discussed here.

The outline is shown below:

- the structure of molecular energy levels (electronic, vibrational and rotational)
- transitions between energy levels, and selection rules (forbidden transitions)
- infra-red spectroscopy and Raman spectroscopy
- molecular fluorescence

Bibliography

Below are some reference books, most of them are available at SCD.

In english

- *Fundamentals of molecular spectroscopy, 4th ed.*, C.N. Banwell and E.M. McCash (McGraw-Hill): best introductory book on the subject, complete without too much maths
- *Physics of atoms and molecules,* Bransden & Joachain (Prentice Hall): chapters 9 and 10.

In french

- *Spectroscopie,* J.M. Hollas (Dunod): meilleur livre disponible en francais (an english version exists but is not available at SCD)
- *Spectroscopie moléculaire,* E. Biémont (De Boek)
- *Spectroscopie dans l'infrarouge*, M. Dalibart & L. Servant (Techniques de l'Ingénieur): bon texte d'introduction (disponible en ligne)
- *Chimie théorique*, Guy Collin, cours en ligne de l'université du Quebec à Chicoutimi (disponible sur http://www.uqac.ca/chimie_ens/Chimie_theorique/)

Chapter 1

Molecular energy levels

In this chapter, we provide a theoretical description of the energy levels of molecules. This description requires quantum mechanics and the use of Schrödinger's equation, similarly to the description of molecular orbitals in physical chemistry.[1] A rigorous quantum mechanical description is demanding and involves some tough algebra. Fortunately, the description of simple molecules only involves a couple of simple models that are analogous to classical systems (the harmonic oscillator and the rigid rotor). As much as possible, we will stick to these simple models.

1.1 Atoms

1.1.1 The hydrogen atom

For atoms, the energy levels are purely electronic states (i.e. orbitals). They can be computed using Schrödinger's equation:

$$\left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{q^2}{r}\right]\Psi(r) = E\Psi(r) \tag{1.1}$$

In the case of hydrogen, this yields

$$E_n = -\frac{E_0}{n^2}\,, \tag{1.2}$$

where $E_0 = -13.6\text{eV}$ (Rydberg energy) and n is the principal quantum number. The eigenfunctions can be separated into a purely radial part and an angular part (spherical harmonics), as:

$$\Psi_{n,l,m} = \frac{1}{r}u_{n,l}(r)Y_l^m(\theta,\phi) \tag{1.3}$$

l is the azimuthal quantum number, m the magnetic quantum number. Each level is degenerated n^2 time.

Definition: an eigenstate is degenerated if for a given eigenvalue (energy) corresponds several eigenfunctions.

[1] see, for instance, CHMA01 lecture

1.1.2 Many electrons atoms

A simple case of many electrons atoms is the case of "hydrogenoids", that is atoms with 1 proton + Z electrons. The energy levels are then given by the following formula:

$$E_n = -\frac{E_0}{n^2} Z^2 \tag{1.4}$$

For more complex cases (many protons + many electrons), Schrödinger's equation becomes difficult to solve exactly. What is important to remember is that the energy levels remain purely electronic levels. Transitions between these levels give spectral lines in the UV/visible range (for example: spectral lamps like Hg, He lamps).

1.2 Born-Oppenheimer approximation

This important hypothesis is crucial to solve Schrödinger's equation in the case of molecules, even the simplest ones (diatomic molecules). Let us decompose the energy of the molecule into its kinetic energy T and its potential energy V.

- kinetic energy $T = T_e + T_n$ (electrons + nucleus)
- potential energy $V = V_{e-n} + V_{e-e} + V_{n-n}$ (electrostatic interaction between electrons and nuclei)

If we consider that nuclei do not move (nuclei frame), $T_n = 0$ and V_{n-n} is constant. Thus, one can define a set of electronic eigenfunctions Ψ_e as:

$$H_e \Psi_e = E_e \Psi_e \tag{1.5}$$

where the electronic Hamiltonian writes $H_e = T_e + V_{e-n} + V_{e-e}$. What we obtain here is the electronic energy and orbitals by solving Schrödinger's equation for electrons only (as in Section 1). But how can we obtain the energy provided by the nuclei?

Born-Oppenheimer approximation: electrons move faster than nuclei, i.e. electrons can react instantly to any nuclear displacement. *Thus, one can consider that the electronic energy E_e is a part of the total potential seen by nuclei.*

$$H_n = T_n + V_{n-n} + E_e$$

H_n is the nuclear Hamiltonian, with $H_n \chi_n = E_n \chi_n$ (χ_n is the nuclear orbital)

The total (molecular) eigenfunction Ψ writes as a product of the *electronic* and *nuclear* orbitals:

$$\Psi = \Psi_e(q, Q) \chi_n(Q), \tag{1.6}$$

where q and Q are the electronic and nuclear coordinates, respectively. Quite similarly, the total energy of the molecule writes as the sum of the energy of electrons and the energy of the nuclei.

$$E_{mol} = E_e + E_n \tag{1.7}$$

The nuclear energy can be again separated into two terms: one term linked to the *rotation* of the nuclei and one term linked to the *vibration* of the nuclei (along the chemical bonds). Finally, the eigenfunctions and total energy writes:

$$\Psi_{mol} = \Psi_e \cdot \Psi_v \cdot \Psi_{rot} \tag{1.8}$$

$$E_{mol} = E_e + E_v + E_{rot} \tag{1.9}$$

Altogether, B-O approximation allows to separate a complex problem into three simpler problems. Furthermore, these three problems are *independent*: for instance the vibrational state of a molecule has no incidence on its rotational energy. We will also demonstrate in the following that $E_e > E_v > E_{rot}$.

Remark: the Born-Oppenheimer approximation remains an approximation. We will see later examples where the B-O approximation breaks down, for instance there is some coupling between the vibrational and rotational energies.

Now let us consider the simplest case of molecules, diatomic molecules.

1.3 Vibration of diatomic molecules

1.3.1 Classical model

The chemical bond is modeled as a spring with stiffness k (rigidity of the bond). This spring models the elongation of the chemical bond due, for example, to thermal excitation. From a classical point of view, this is an harmonic oscillator: two masses m_1 and m_2 linked with a spring. Let us note μ the reduced mass, defined as:

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} \tag{1.10}$$

If $x(t)$ is the elongation of the bond, the equation of motion is $\mu \frac{d^2x}{dt^2} = -kx$, or:

$$\ddot{x} + \omega_0^2 x = 0; \text{ where } \omega_0^2 = \frac{k}{\mu} \tag{1.11}$$

The solution of this differential equation is known: $x(t) = x_0 \cos(\omega_0 t + \varphi)$ where x_0 is the maximum amplitude of the oscillation.

Remark: this corresponds to a motion inside an harmonic potential $V = \frac{1}{2}kx^2$

Remark 2: the value of the stiffness constant k depends on the strength of the chemical bond. Typical values are $k \simeq 500$ N.m^{-1} for a single covalent bond, $k \simeq 1000$ N.m^{-1} for a double bond and $k \simeq 1500$ N.m^{-1} for a triple bond.

1.3.2 Quantum model

In the frame of quantum mechanics, it is necessary to solve Schrödinger's equation:

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} + \frac{1}{2}\mu\omega^2 x^2 \right] \Psi(x) = E_v \Psi(x) \tag{1.12}$$

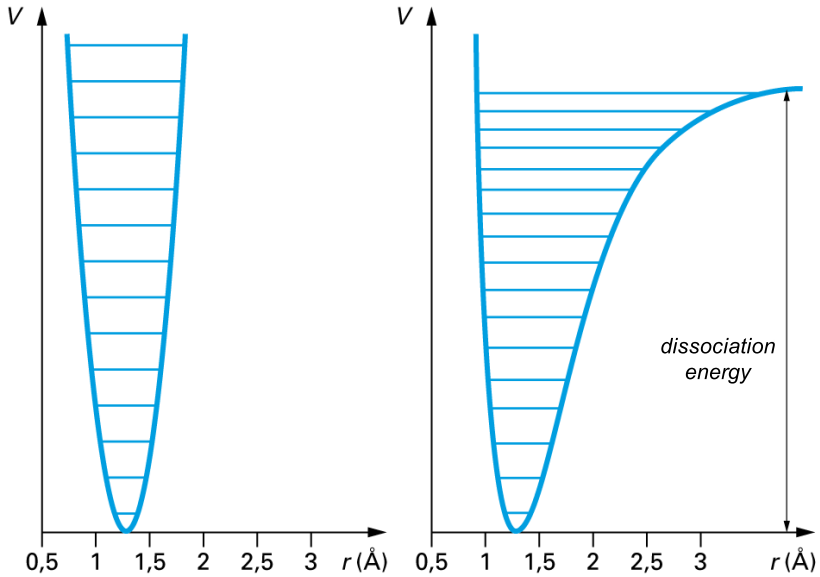

Figure 1.1: *The vibration potential of a diatomic molecule. Left panel: harmonic oscillator. Right panel: Morse potential.*

where E_v stands for the vibrational energy. Solving this equation is outside the scope of this lecture. The important result is that *the vibrational energy is quantified* and writes:

$$E_v = (v + \frac{1}{2})h\nu_0 = (v + \frac{1}{2})\hbar\omega_0 \tag{1.13}$$

where v is the vibrational quantum number $v = 0, 1, 2, ...$ The energy levels are equidistant and $\hbar\omega_0$ apart (see Fig. 1.1). Alike to the classical harmonic oscillator, the angular frequency is $\omega_0 = \sqrt{\frac{k}{\mu}}$ (SI units rad.s^{-1}) and the frequency is $\nu_0 = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$ (SI units: Hz).

1.3.3 Anharmonic oscillator

Real molecules do not abide by the laws of simple harmonic motion. For instance, if the atoms are stretched too far from each other, the chemical bond will eventually break. This means that the potential should vanish when $x \to +\infty$. In contrast, when the atoms become too close, the interpenetration of their electron clouds leads to a strong repulsion. A good choice to model the molecular interaction is the *Morse potential* (see Fig. 1.1, right panel).

$$V(r) = D_e \left(1 - e^{-\beta(r-r_e)}\right)^2 \tag{1.14}$$

In this equation, r_e is the equilibrium position, D_e is the dissociation energy, and β is an interaction constant. All these parameters depend on the considered molecule. When this potential is introduced into Schrödinger's equation, one obtains for the energy:

$$E_v = (v + \frac{1}{2})\hbar\omega_e - (v + \frac{1}{2})^2\hbar\omega_e\chi_e + ... \tag{1.15}$$

where ω_e is a new angular frenquency and χ_e is the anharmonicity constant which is always small and positive (typically $\chi_e \simeq 0.01$). It is clear from the previous equation that the energy levels are no longer equidistant, but become closer when v increases.

1.4 Rotation of diatomic molecules

1.4.1 Classical model: the rigid rotor

In this model the chemical bond is rigid. In other words, masses m_1 and m_2 are joined by a rigid bar whose length is $r_e = r_1 + r_2$ (this length corresponds to the bond equilibrium length). The molecule rotates about its center of mass, which is taken as the frame origin O. The center of mass is defined by the balancing equation $m_1 r_1 = m_2 r_2$. To solve this problem it is simpler to use the concept of moment of inertia I. This quantity is a measure of an object's resistance to changes to its rotation (SI units kg·m^2). It is defined as follow:

$$I = \sum_i m_i r_i^2 \tag{1.16}$$

where r_i is the distance of the mass to the rotation axis. In the case of the rigid rotor, the moment of inertia writes

$$I = m_1 r_1^2 + m_2 r_2^2 = \mu r_e^2 \tag{1.17}$$

Equation 1.17 means that our problem is equivalent to a single fictitious mass μ rotating at a distance r_e from the origin with angular frequency ω. The total energy of the system is purely kinetic and writes:

$$E_{rot} = \frac{1}{2}\mu v^2 = \frac{1}{2}\mu r_e^2 \omega^2 = \frac{1}{2} I \omega^2 \tag{1.18}$$

Description using the angular momentum $\vec{L}$: by definition

$$\overrightarrow{L} = \overrightarrow{OM} \wedge \overrightarrow{p} = \overrightarrow{OM_1} \wedge \overrightarrow{p_1} + \overrightarrow{OM_2} \wedge \overrightarrow{p_2} = r_1 \overrightarrow{u_r} \wedge m_1 r_1 \omega \overrightarrow{u_\theta} + r_2 \overrightarrow{u_r} \wedge m_2 r_2 \omega \overrightarrow{u_\theta} \tag{1.19}$$

and taking the modulus

$$||\overrightarrow{L}|| = m_1 r_1^2 \omega + m_2 r_2^2 \omega = I\omega = \mu r_e^2 \omega \tag{1.20}$$

We finally obtain the following alternative expressions for the total energy:

$$E_{rot} = \frac{1}{2} I \omega^2 = \frac{L^2}{2I} = \frac{L^2}{2\mu r_e^2} \tag{1.21}$$

1.4.2 Quantification

In quantum mechanics, the squared modulus of the angular momentum is quantified: $L^2 = J(J+1)\hbar^2$ where $J = 0, 1, 2, ...$ is the rotational quantum number. As a consequence, the rotational energy is also quantified:

$$E_{rot} = BJ(J+1), \text{ where } B = \frac{\hbar^2}{2\mu r_e^2} = \frac{\hbar^2}{2I} \tag{1.22}$$

B is known as *the rotational constant* and has the dimension of an energy. The rotational energy levels are represented on Fig. 1.2. In contrast with vibrational (harmonic) levels, they are not equidistant.

Remark: each rotational level is $(2J+1)$ degenerated.

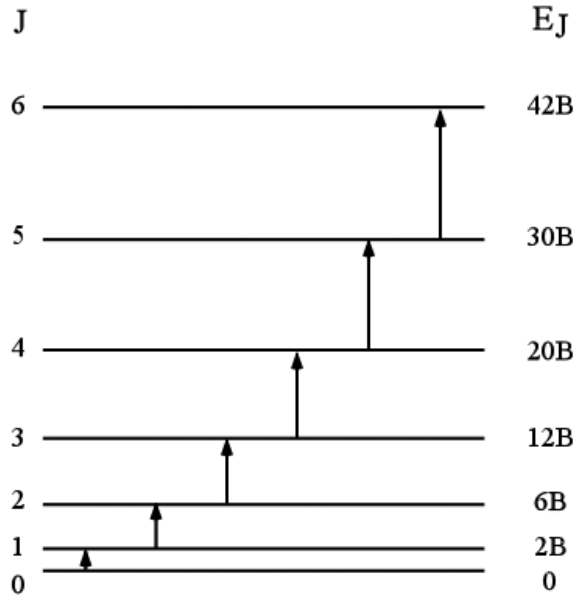

Figure 1.2: *The rotational energy levels of a diatomic molecule as a function of the rotational quantum number J.*

1.4.3 Non-rigid rotor

All chemical bonds are actually elastic to some extent and the bond length increases with J. The rigid rotor model can thus be refined into the non-rigid rotor model. This yields for the expression of the rotational energy:

$$E_{rot}(J) = BJ(J+1) - DJ^2(J+1)^2 + ... \tag{1.23}$$

where the centrifugal distorsion constant is $D = \dfrac{h^3}{32\pi^4 I^2 r_e^2 k}$.
The correction term $DJ^2(J+1)^2$ is almost negligible, and takes a significant value only for high J values ($J > 10$).

1.5 Polyatomic molecules

1.5.1 Vibration of polyatomic molecules

Let us consider a molecule consisting of N atoms. Each atom has 3 degrees of freedom, thus the molecule has $3N$ degrees of freedom. But among them, 3 degrees of freedom are bare translations of the whole molecule and 3 others are rotations. This leaves $3N - 6$ fundamentals vibrations for a molecule with N atoms. An important exception is the case of *linear molecules*, where there is no rotation about the molecule axis. Hence only two degrees of rotational freedom are required, leaving $3N - 5$ fundamental vibrations. In brief:

- non-linear molecules with N atoms: $3N - 6$ fundamental vibrations
- linear molecules with N atoms: $3N - 5$ fundamental vibrations

Normal modes of vibration

The normal modes of vibrations are the mathematical frame describing the vibrations of polyatomic molecules. Let us start with the energy of the harmonic oscillator:

$$E = E_c + E_p = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 = \frac{1}{2}(m\dot{x}^2 + m\omega_0^2 x^2) \tag{1.24}$$

Then we introduce the new variable $q = \sqrt{m}x$ and we substitute into Eq. 1.24:

$$E = \frac{1}{2}(\dot{q}^2 + \omega_0^2 q^2) \tag{1.25}$$

Equation 1.25 gives the energy of a single harmonic oscillator with angular frequency ω_0. Now, in the case of a N-atoms molecule, the total vibrational energy of the nuclei H can always be written using the normal coordinates Q_k as:

$$H = \frac{1}{2}\sum_{k=1}^{3N-6}\left(\dot{Q}_k^2 + \omega_k^2 Q_k^2\right) \tag{1.26}$$

This energy corresponds to $3N-6$ (or $3N-5$ in the case of linear molecules) independent harmonic oscillators with angular frequencies ω_k. There are $3N-6$ independent equations of motion admitting $3N-6$ solutions:

$$\sum_{k=1}^{3N-6}\ddot{Q}_k + \omega_k^2 Q_k = 0 \tag{1.27}$$

$$Q_k = Q_{k0}\cos(\omega_k t) \tag{1.28}$$

These solutions Q_k are called normal modes of vibration. Each normal mode is characterized by its angular frequency ω_k and induces the motion of all nuclei. The vibrational energy writes using the normal modes:

$$E_v = \sum_{k=1}^{3N-6}\hbar\omega_k\left(v_k + \frac{1}{2}\right) \tag{1.29}$$

In the previous expression, degenerated modes contribute only once (i.e., if a mode is doubly degenerated, do not count it twice in the energy).

1.5.2 Rotation of polyatomic molecules

For polyatomic molecules, there are in general three different inertia momenta: $I_A \neq I_B \neq I_C$. Hence, the rotational energy writes:

$$E_{rot} = AL_x^2 + BL_y^2 + CL_z^2 \tag{1.30}$$

where A, B, C are form factors, with for instance $A = \dfrac{\hbar^2}{2I_A}$. Let us consider different "molecular shapes".

Linear molecule: $I_B = I_C$; $I_A = 0$. The energy has the same expression as a diatomic molecule: $E_{rot} = BJ(J+1)$
However, the molecule being heavier than its diatomic counterpart, B is lower yielding to lower rotational energies.
Sample molecules: CO_2, HCN...

Spherical molecule: $I_A = I_B = I_C$ and $L^2 = L_x^2 + L_y^2 + L_z^2$. Same expression for the rotational energy: $E_{rot} = BJ(J+1)$
Sample molecules: CCl_4, CH_4...

Symmetric tops: two identical inertia momenta. For historical reasons, two cases are considered.

- "oblate" molecules ("Smarties" shaped) with $I_A = I_B > I_C$ and rotational energy

$$E_{rot} = BJ(J+1) + (C-B)K^2 \text{ where } -J \leq K \leq J$$

- "prolate" molecules (rugby ball or cigar-shaped) where $I_A < I_B = I_C$ and rotational energy

$$E_{rot} = BJ(J+1) + (A-B)K^2$$

Sample molecules: NH_3, C_6H_6 (oblates), CH_3Cl (prolate)

Asymmetric tops: where $I_A \neq I_B \neq I_C$. No general expression!

Chapter 2

Transitions between energy levels

Spectroscopic techniques probe the molecular energy levels through *transitions* between those levels. In other words, the levels $E(n, v, J)$ are obtained thanks to the measurement of various ΔE

- either by absorption, emission or scattering of photons, *(scope of this lecture)*
- or by absorption, emission or scattering of other particles such as electrons, ions...

The main use of spectroscopy is the identification of molecules *via* their spectrum and the so-called "molecular fingerprints" (spectral lines characteristic of a given molecule). However, energy levels are also sensitive to the environment of the molecule (for instance electric field, temperature, pH...). Hence, spectroscopy can also be used as a diagnostic tool to investigate the physical and chemical conditions inside a medium.

2.1 Main experimental schemes in optical spectroscopies

2.1.1 Absorption spectroscopy

Here the sample is irradiated by a broadband source ("white light"), and the attenuation of this light is recorded. One obtains a set of absorption lines. The resulting spectrum $I(\lambda)$ is generally normalized by the reference spectrum $I_0(\lambda)$ and one plots the absorbance A defined as:

$$A(\lambda) = -\log \frac{I}{I_0} \tag{2.1}$$

The absorbance depends on the absorbing properties of the sample. In particular, in the case of liquids (molecules in a solution), the absorbance depends on the molecular concentration C according to Beer-Lambert's law:

$$A(\lambda) = \varepsilon(\lambda) \cdot C \cdot \ell \tag{2.2}$$

where ℓ is the sample length (units: cm) and ϵ is the molar absorption coefficient (units: L.mol^{-1}.cm^{-1}).

2.1.2 Emission spectroscopy

Here the radiation emitted by the sample is directly analyzed. Obviously, the sample must be in an excited state. Several methods of excitation are possible, depending on the type of sample considered:

- electric discharge for a gas
- electroluminescence for a semiconductor
- heating (samples absorbing in the far infrared range)
- chemical excitation: chemiluminescence
- optical excitation (fluorescence and phosphorescence)

2.1.3 Scattering spectroscopy

The molecule is irradiated by a radiation whose frequency does not match any of the existing molecular transitions *(non-resonant excitation)*. Two cases can be distinguished:

- **Elastic scattering**, or Rayleigh scattering: the incident radiation is scattered (i.e. its direction changes) without any changes in its frequency.
- **Inelastic scattering**, or Raman scattering: the energy (frequency) of the scattered photons ν_d is different from the energy of the incident photons ν_0, and there is a change of the molecule energy. The difference $\nu_d - \nu_0$ match a molecular transition.

2.2 Position, width and intensity of spectral lines

2.2.1 Position

A spectral line position ν is directly given by the transition energy:

$$\Delta E = E_f - E_i = h\nu \tag{2.3}$$

where ν is the frequency of the photon being absorbed, emitted, or scattered.

2.2.2 Linewidth

Spectral lines are not Dirac functions, but exhibit a certain width which is called *linewidth*. The linewidth has several origins, some practical (instrumental limitations), some fundamental.

Instrumental limitations

Any spectrograph has a limited spectral resolution, meaning that it is not always able to separate λ from $\lambda + d\lambda$. In a diffractive spectrograph, the resolution mainly depends on the exit slit width, and the diffractive grating pitch.

Fundamental limitations

Even with a perfect instrument, a given spectra line present a certain width. The main broadening causes are the following.

- **pressure broadening:** collisions between nearby particles disturb the electron cloud and the molecular vibrations. This effect is particularly important in liquids.
- **Doppler broadening:** since molecules are in motion, their absorption or emission is shifted due to the Doppler effect. Brownian motion is random, thus the Doppler effect equally shifts towards higher and lower frequencies. This kind of broadening is important for gases.
- **Uncertainty principle broadening:** even an isolated and fixed molecule does not exhibit infinitely thin spectral lines, because Heisenberg's uncertainty relations somehow "blur" the position of the energy levels. If Δt is the excited level lifetime, then this energy level presents a energy width ΔE as:

$$\Delta E \cdot \Delta t \approx \hbar \tag{2.4}$$

  ΔE is called the natural linewidth of this spectral line. It can be estimated from the following relations:

$$\Delta \nu = \frac{\Delta E}{h} \approx \frac{h}{2\pi h \Delta t} \approx \frac{1}{2\pi \Delta t} \tag{2.5}$$

  For optical spectroscopy, the natural linewidth is very small compared to the involved frequencies.

2.2.3 Intensity

For a transition between two energy levels E_i and E_f, the intensity of the associated spectral line depends mainly on three parameters.

- **the length of the sample:** as in Beer-Lambert's law
- **the transition probability:** varies between 0 and 1, and can be calculated using quantum mechanics. Simply speaking, it depends on the overlapping integral of the two associated wavefunctions. In some cases, the transition probability is null, the corresponding transition is called a *forbidden transition*.
- **the population of the initial state:** a spectral line will be more intense if an important number of molecules is able to undergo the corresponding transition, that is to say, a great number of molecules is in the initial state E_i. The proportion of molecules in the E_i state is called the population of the E_i state. At thermodynamic equilibrium, the population n_i of the energy level i is governed by the Boltzmann distribution:

$$n_i = n_0 \exp\left(-\frac{E_i - E_0}{kT}\right) \tag{2.6}$$

  n_0 being the population of the fundamental level, and k is the Boltzmann constant. Hence, the lower energy levels are the most populated, and higher levels becomes populated when the temperature increases. Note that the following equation is only valid for non-degenerated energy levels. If the levels are degenerated, the Boltzmann factor must be multiplied by the degeneracy factor (see tutorial for an example).

2.3 Selection rules

In a molecule, the possible transitions between energy levels are not all allowed. Some transitions are forbidden, either because the transition probability is null (or extremely weak), or because interaction with an impinging electromagnetic wave is not possible. The set of allowed and forbidden transitions is called the *selection rules* of a given molecule.

2.3.1 Selection rules from transition probabilities

For some transitions, quantum mechanics predict a transition probability equals to zero. These selections rules depend on the kind of involved energy levels (electronic, vibrational...).
Example: transitions between two electronic levels $n \to n'$ are only permitted with the following rules $\Delta l = \pm 1$ and $\Delta m = \pm 0, 1$. Transitions with $\Delta l = 0$ are forbidden.

Calculation of the selection rules is generally complex. In the scope of this lecture we will only state the selection rules without any demonstration.

2.3.2 Selection rules from the interaction with light

Apart from the transition probabilities, a necessary condition is that the molecule can interact with light in order to absorb or emit photons. Hence, there must be some electric or magnetic effect produced by the change of energy level which can be influenced by the electric or magnetic fields associated with light. Such a molecule will said to be "active" (Raman-active, IR-active, microwave-active...)

Chapter 3

Vibrational spectroscopy

In this section, we will discuss the two spectroscopic techniques probing the vibrational energy levels: infrared spectroscopy and Raman spectroscopy.

3.1 Classical theory of Raman scattering

Let us consider a diatomic molecule with a single fundamental mode of vibration Q_k.

$$Q_k = Q_{k_0} \cos(\omega_k t) \tag{3.1}$$

This molecule is illuminated with monochromatic light, such as a laser beam. Its response to the exciting electromagnetic field is governed by the electric dipole moment $\overrightarrow{\mu}$, which presents two contributions:

$$\overrightarrow{\mu} = \overrightarrow{\mu}_p + \overrightarrow{\mu}_{ind} \tag{3.2}$$

$\overrightarrow{\mu}_p$ is the permanent dipole moment, arising from the different electronegativies of the atoms (heteropolar molecules). $\overrightarrow{\mu}_{ind}$ is the induced dipole moment: under the influence of the incident electromagnetic field the electron cloud distorts, inducing a dipole moment. The induced moment is generally expressed as a function of α, the *polarisability* of the molecule:

$$\overrightarrow{\mu}_{ind} = \alpha \overrightarrow{E_L} = \alpha \overrightarrow{E_0} \cos(\omega_L t)\,, \tag{3.3}$$

where $\overrightarrow{E_L}$ is the incident electric field (laser field), and ω_L is the laser angular frequency. The polarisability quantifies the susceptibility of the electron cloud to distort under the influence of the laser field: the higher the polarisability, the larger the deformation of the electron cloud.

Considering the vibration mode Q_k, it is possible to develop α and $\overrightarrow{\mu}_p$ into their first order Taylor series:

$$\overrightarrow{\mu}_p = \overrightarrow{\mu_{p_0}} + Q_k \left(\frac{\partial \overrightarrow{\mu_p}}{\partial Q_k} \right)_{Q_k=0} \tag{3.4}$$

where $\overrightarrow{\mu_{p_0}}$ is the permanent dipole moment *in the absence of vibration.* Similarly, the polarisability writes:

$$\alpha = \alpha_0 + Q_k \left(\frac{\partial \alpha}{\partial Q_k} \right)_{Q_k=0} \tag{3.5}$$

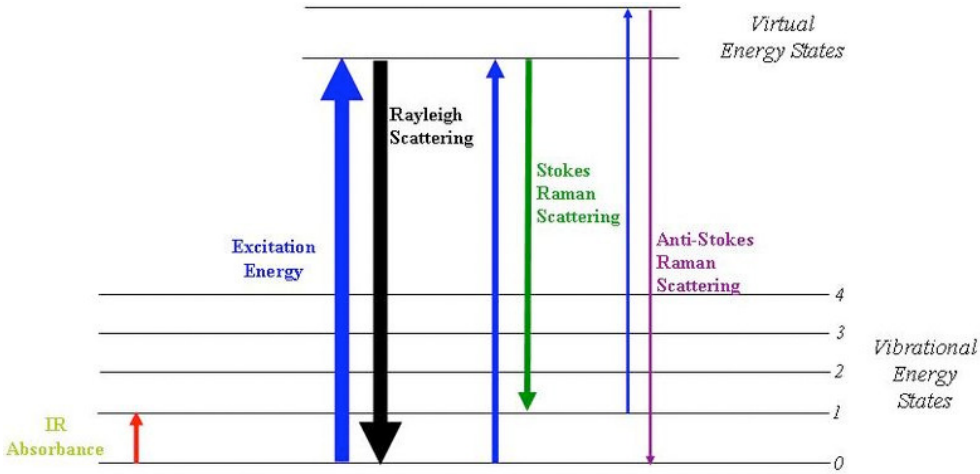

Figure 3.1: *Energy diagram showing the different scattering processes in a molecule.*

This yields:

$$\overrightarrow{\mu} = \overrightarrow{\mu_{p0}} + \left(\frac{\partial \overrightarrow{\mu_p}}{\partial Q_k}\right)_0 Q_{k_0}\cos(\omega_k t) + \alpha_0 \overrightarrow{E_0}\cos(\omega_L t) + \left(\frac{\partial \alpha}{\partial Q_k}\right)_0 \overrightarrow{E_0}\cos(\omega_L t) Q_{k_0}\cos(\omega_k t) \tag{3.6}$$

and developping the last term:

$$\begin{aligned}\overrightarrow{\mu} &= \overrightarrow{\mu_{p0}} + \left(\frac{\partial \overrightarrow{\mu_p}}{\partial Q_k}\right)_0 Q_{k_0}\cos(\omega_k t) + \alpha_0 \overrightarrow{E_0}\cos(\omega_L t) \\ &+ \frac{1}{2}\left(\frac{\partial \alpha}{\partial Q_k}\right)_0 Q_{k_0}\overrightarrow{E_0}\Big[\cos\big((\omega_L - \omega_k)t\big) + \cos\big((\omega_L + \omega_k)t\big)\Big]\end{aligned} \tag{3.7}$$

It clearly appears from the previous equation that the total dipole moment is the sum of four terms, corresponding to four dipoles oscillating at four different frequencies.

- ω_k: infrared absorption at the molecular vibration frequency
- ω_L: Rayleigh scattering (elastic scattering of the laser light)
- $\omega_L \pm \omega_k$: Raman scattering (inelastic scattering). The term at frequency $\omega_L - \omega_k$ is called Stokes scattering, the term at $\omega_L + \omega_k$ is the anti-Stokes scattering.

These different scattering processes are illustrated in Fig. 3.1. Raman and Rayleigh scattering are described thanks to a "virtual" energy level, which does not correspond to any real energy state of the molecule. This virtual state is only introduced as a convenient description of the scattering process. After Rayleigh scattering the molecule remains in the same energy level, while after Raman scattering the molecule undergoes a change in its energy by an amount ω_k.

From equation 3.7 it is possible to deduce the **vibrational selection rules**:

- a vibration mode is **Raman-active** if $\left(\frac{\partial \alpha}{\partial Q_k}\right)_{Q_{k_0}} \neq 0$. The physical meaning of this mathematical condition is the following: the electron cloud should distort under the

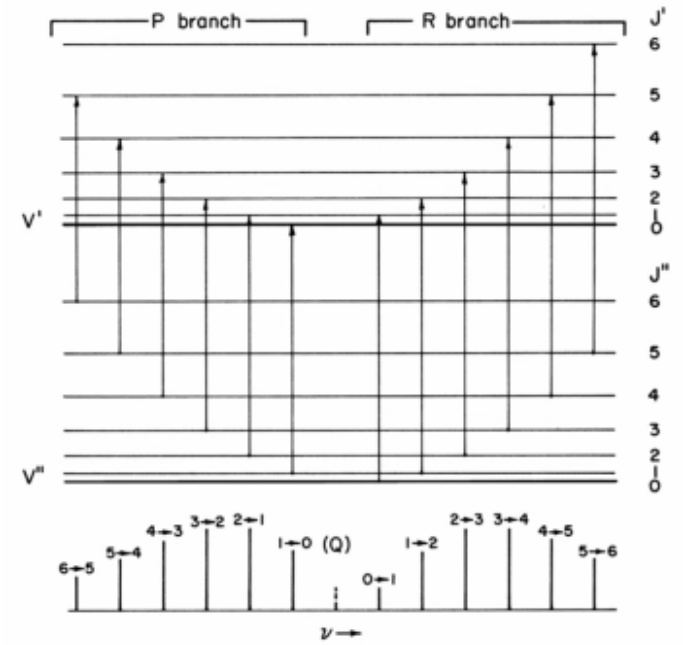

Figure 3.2: *Upper panel: energy diagram showing infrared absorption in a diatomic molecule: the only allowed transitions are P and R branches. Transitions with $\Delta J = 0$ (Q branch) are forbidden. Lower panel: schematic IR spectra deduced from the energy diagram.*

molecular vibration. In other words, the molecular volume must change of shape, orientation or size when the molecule vibrates.

- a vibration mode is **infrared-active** if $\left(\frac{\partial \overrightarrow{\mu_p}}{\partial Q_k}\right)_{Q_{k_0}} \neq 0$. This means that the vibration must induce some kind of variation of the permanent dipolar moment.

Important: in the case of polyatomic molecules, this selection rules apply *for each fundamental mode of vibration.* For a given molecule, some modes can be active, and others not.

3.2 Rovibrational spectroscopy of diatomic molecules

3.2.1 Infrared absorption

For a heteropolar molecule (IR-active), the allowed transitions are:

$$\Delta v = \pm 1 \text{ and } \Delta J = \pm 1 \tag{3.8}$$

In other words, any change in the vibrational state induces a change in the rotational state (transitions with $\Delta J = 0$ are forbidden). The rovibrational transition are usually named (see Fig. 3.2):

- P branch: transitions with $\Delta J = -1$
- R branch: transitions with $\Delta J = +1$

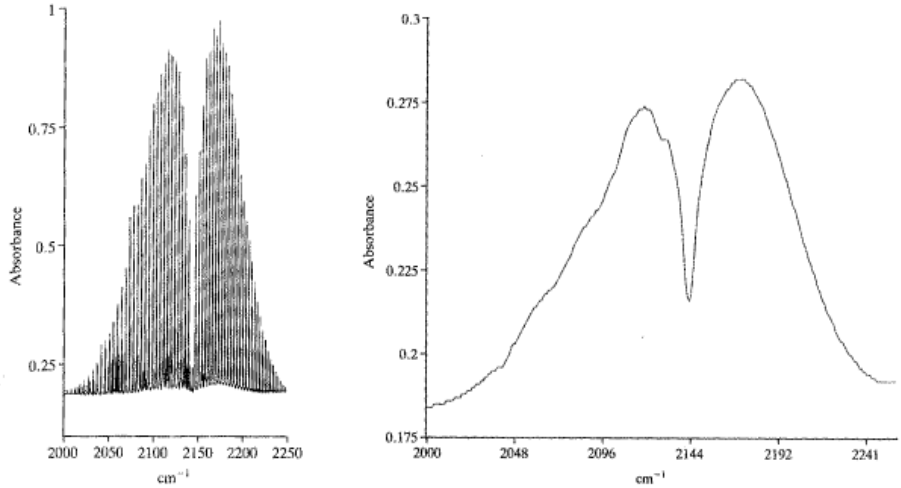

Figure 3.3: *Two infrared spectra of the CO molecule. Left panel: high resolution spectrum showing the fine rotational spectrum. Right panel: low resolution spectrum, only the contour of the P and R branches is resolved.*

Remark 1: The fine rotational structure of a rovibrational spectrum can only be seen with a high-resolution spectrograph (since the rotational lines are separated from each other by a few cm^{-1}). Moreover, the molecules should be in a gaseous phase and at low pressure (otherwise, collisions between molecules blur the rotational spectrum). At low resolution, only the contour of the spectra is seen (see Fig. 3.3)

Remark 2: in the frame of Born-Oppenheimer approximation, there is no coupling between the vibrational and rotational states, hence the rotational constant B is the same for the energy levels $v = 0$ and $v = 1$. In fact, there is a small rovibrational coupling and B decreases a bit when v increases (because, simply speaking, the length of the chemical bond increases with the vibrational quantum number). For instance, for the CO molecule one observes $B_0 = 1.915$ cm^{-1} and $B_1 = 1.989$ cm^{-1}.

Remark 3: transitions with $\Delta v = \pm 1, \pm 2, ...$ are also allowed. They are called *overtones* ($\Delta v = \pm 1$ is the first overtone, $\Delta v = \pm 2$ the second overtone, etc). However, overtones exhibit a considerably lesser intensity than the fundamental vibrational transition.

3.2.2 Raman spectroscopy

The selection rules for Raman spectroscopy are, for the vibrational part:

$$\Delta v = +1 \text{ (Stokes) and } \Delta v = -1 \text{ (Anti} - \text{Stokes)} \tag{3.9}$$

and for the rotational part, the selection rules are:

- $\Delta J = 0$: Q branch, giving rise to one spectral line
- $\Delta J = \pm 2$: O branch ($\Delta J = -2$) and S branch ($\Delta J = +2$), giving rise to a spectrum of lines $4B$ apart.

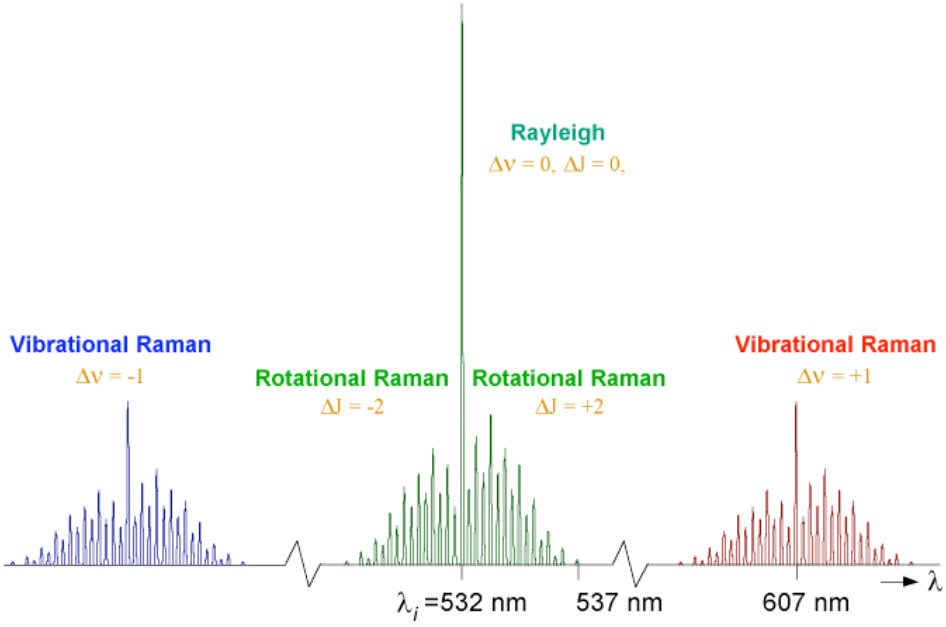

Figure 3.4: *Raman spectrum of the N_2 molecule, recorded using a laser source at $\lambda = 532$ nm. The Rayleigh, Stokes and anti-Stokes contributions are shown.*

Transitions with $\Delta v = 0$ and $\Delta J = \pm 2$ give rises to a set of lines around the Rayleigh line which is called *rotational Raman* (since the vibrational state of the molecule does not change). See Fig. 3.4 for an example of Raman spectra showing these different lines. However, in most experimental schemes, the rotational substructure of the spectrum is not resolved and only the Q branch is seen.

Important remark: Raman spectra are often represented with the so-called "Raman shift" as an energy unit. The Raman shift is defined as the energy difference between the Raman lines (the Stokes line in general) and the Rayleigh line (i.e. the laser frequency). In other words, even if the exciting laser is in the visible range, the Raman shift is in the infrared range. This allows direct comparison between the IR absorption spectrum and the Raman spectrum.

Intensity of Raman lines

It is important to notice that the Stokes and anti-Stokes Raman lines exhibit very different intensities. Anti-Stokes scattering ($\Delta v = -1$) takes places from an excited vibrational level, which is thermally less populated than the fundamental level. Hence, the Stokes line is considerably more intense than the anti-Stokes line. It is possible to show that the respective intensities of the Stokes (I^S) and anti-Stokes (I^{AS}) lines are:

$$\frac{I^{AS}}{I^S}(k) = \left(\frac{\omega_L + \omega_k}{\omega_L - \omega_k}\right)^4 \exp\left(-\frac{\hbar\omega_k}{kT}\right) \tag{3.10}$$

Thus, for a given mode k and a given temperature T, one has $I_k^{AS} \ll I_k^S$.

3.3 Polyatomic molecules

In this case, it is not possible to determine general selection rules. The selection rules depends on the symmetry of the molecule and are determined using group theory. Each molecule belongs to a symmetry group which imposes selection rules. In the following, we will see some examples.

3.3.1 Linear molecules

For the IR absorption spectroscopy of linear molecules, the selection rules depends on the kind of vibration

- vibration parallel to the molecular axis: $\Delta v = \pm 1$ and $\Delta J = \pm 1$ (P and R branches)
- vibration perpendicular to the molecular axis: $\Delta v = \pm 1$ and $\Delta J = 0; \pm 1$ (branches Q, P and R)

Example: HCN molecule

3.3.2 Skeleton vibrations

Such modes exist in "big" molecules, and arise from linear (such as $C - O - C$) or crossed chains of atoms inside molecules, or from aromatic rings. These modes give rise to absorption bands (often between 700 and 1400 cm^{-1}) that are easily identified in a spectrum.

3.3.3 Group vibrations

Group vibrations correspond to vibrations of a light molecular termination attached to a bigger molecule. If the core molecule is heavy enough, one obtains characteristic vibration lines.
Examples: terminations such as $-CH_3$, $-OH$, or $-CO$.

There are data bases collecting these characteristic bands, allowing the experimentalist to identify molecules.

3.4 Raman spectroscopy of solids

One of the strengths of Raman spectroscopy lies in the fact that this technique can be easily applied to the spectroscopy of solids. In a crystal, there is N atoms, where N is a huge number. The number of allowed vibrations is hence potentially huge, but it is sufficient to look at the vibrations inside the primitive crystalline cell. These fundamental vibrations are *phonons* – quantified vibrations inside a crystal. In other words, Raman scattering in a crystal takes place through the absorption or emission of a phonon. In the case of solids, Raman spectroscopy allows to probe crystalline structures. For instance, it allows to discriminate between different forms of solid carbon (graphite, diamond, charcoal...)

3.5 Instrumentation

In the infrared range, most experimental schemes use Fourier Transform Spectroscopy, or FTIR spectroscopy. The principle of FT spectroscopy is to record an interferogram containing all the spectral information in a single figure. Then, the spectrum is retrieved by means of a mathematical operation called a Fourier Transform. In practice, a FTIR spectrograph includes a Michelson interferometer with a mobile mirror. It is the precise displacement of this mirror that allows to record the interferogram.

FTIR spectroscopy allows to record broadband spectra much more quickly and with a better signal-to-noise ratio than diffractive (grating) spectrographs.

Chapter 4

Molecular fluorescence

Here, we present a survey of the fluorescence phenomenon, using simple models. There are a number of different words associated with this phenomenon, sometimes thought to be synonyms. It is therefore useful to start with some vocabulary. *Luminescence* is the generic term to describe the non-thermal emission of light by matter. The word *photoluminescence* covers the case where matter is optically excited, and *fluorescence* is a special case of photoluminescence, as we shall see below. Below, we will focus on the case of *molecular fluorescence*, that is to say the fluorescence emission from a (generally organic) molecule. The case of solid-state emitters, either bulk solids or quantum dots, can be described using a very similar frame and will be discussed in the last section.

4.1 Description of electronic states

Fluorescence involves transitions between *electronic* states, and usually gives birth to fluorescence emission in the UV/visible range. Fluorescent molecules are generally complex organic molecules presenting aromatic rings. These rings contain delocalized electrons, allowing interaction between the molecule and light. To describe the fluorescence phenomenon, it is generally sufficient to use a simplified description of the electronic energy levels, involving only the fundamental and excited states. The lowest energy electronic transition appears between the highest occupied molecular orbital (or HOMO) and the

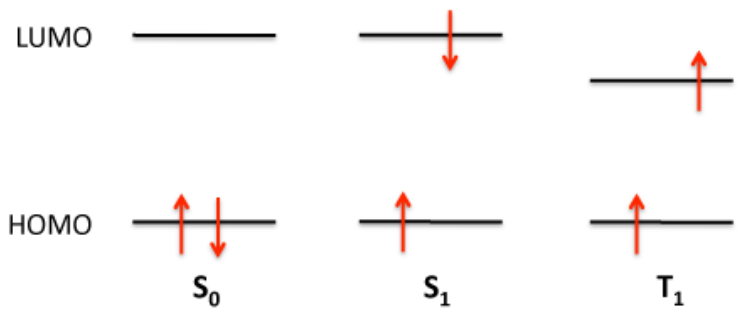

Figure 4.1: *Singlet and triplet states: states S_0 and S_1 are the fundamental and excited singlet states (with $s_1 = \frac{1}{2}$ and $s_2 = -\frac{1}{2}$). State T_1 is the first triplet state, where both electron spins point towards the same direction. Note the lower energy of the triplet state than the equivalent singlet state.*

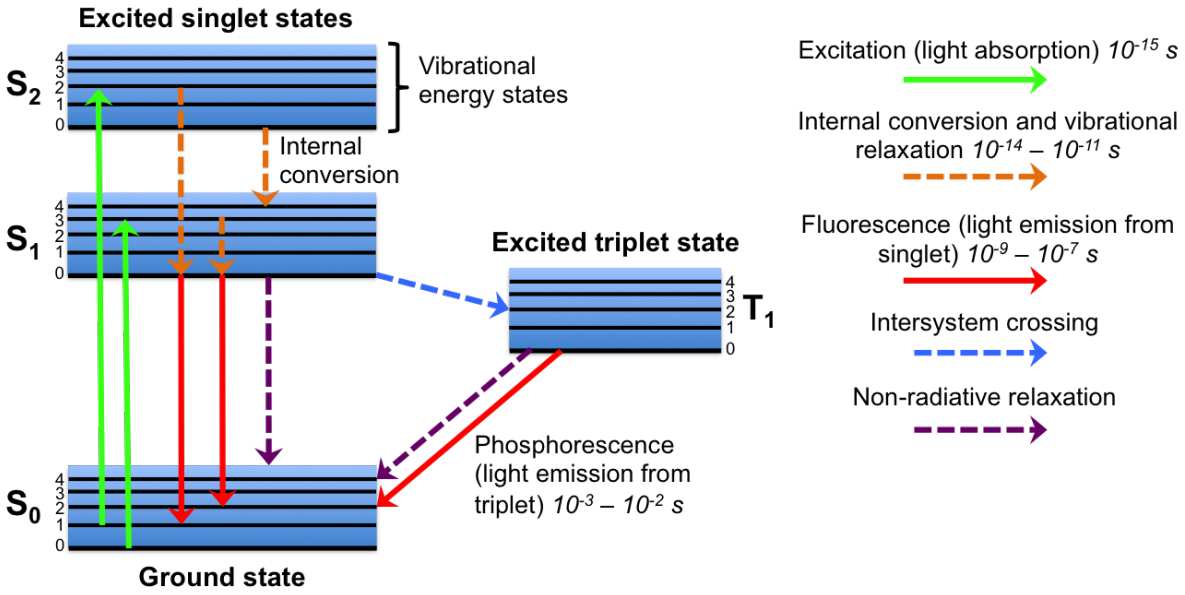

Figure 4.2: *Jablonski energy diagram, showing the different relaxation processes inside an organic molecule with their characteristic decay times.*

lowest unoccupied molecular orbital (LUMO). Hence, the fluorescence emission corresponds to a LUMO $\rightarrow$ HOMO radiative transition. Higher energy levels have to be taken into account to describe higher energy fluorescent lines.

Moreover, molecular orbitals are labeled according to the multiplicity of their electronic spin $2S+1$. In this expression S is the total spin momentum, corresponding to the sum of each electron spin: $S = s_1 + s_2$, where $s_i = \pm\frac{1}{2}$ is the spin of an individual electron. States with a multiplicity $2S+1 = 1$ are called *singlet states*, while states with $2S+1 = 3$ are called *triplet states*. See Fig. 4.1 for an illustration of these electronic states.

In principle, when a photon is absorbed or emitted, the total spin is conserved. This yields the following selection rule: transitions S$\leftrightarrow$T are forbidden. However, the presence of the spin-orbit coupling (interaction between the electron spin and its motion, that is, interaction between operators L and S) relaxes this condition, and allows transitions between singlet and triplet states, although with a very low probability. Such transitions are called *intersystem crossing*. These energy levels are often represented on a Jablonski diagram (Fig. 4.2), where the electronic states are filed according to their multiplicity.

In Fig. 4.2, the different relaxation processes are shown with their characteristic decay times. For instance, since the $T_1 \rightarrow S_0$ transition has a very low probability, the corresponding decay time is high (in the ms range). This transition gives rise to the phenomenon of *phosphorescence*. Transitions between excited singlet and triplet states, enabled by the spin-orbit coupling, are called *intersystem crossing*. The relaxation from a given energy state to a lower energy state can be either radiative (i.e. involving the emission of a photon) or non-radiative. In the above Jablonski diagram, radiative processes are shown as solid lines and non-radiative processes as dashed lines. Non-radiative processes, such as internal conversion or vibrational relaxation, generally involve a relaxation via the vibrational states and the emission of heat. *Vibrational relaxation* is the relaxation of the molecule down the vibrational level ladder of a given electronic state. In contrast,

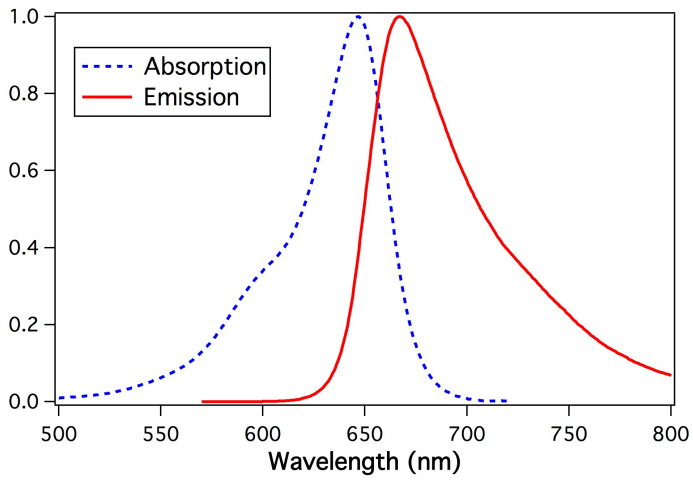

Figure 4.3: *Normalized absorption and fluorescence emission spectra of the ATTO 647 molecule. Spectroscopic data from the manufacturer.*

internal conversion involves a non-radiative transition, via the vibrational levels, towards another electronic state of the same multiplicity.

4.2 Fluorescence spectra of molecules

Fig. 4.3 shows a representative room temperature fluorescence spectrum for an organic dye molecule, here ATTO 647, a common molecule absorbing and emitting in the red part of the visible spectrum. Several features appear on this spectrum. First, the emission spectrum is red-shifted with respect to the absorption spectrum. This energy difference is called the *Stokes shift*. Second, the main absorption and emission peaks exhibit an asymmetric lineshape and are mirror images of each other. All these features can be qualitatively explained with the help of the Jablonski diagram.

At room temperature, the thermal energy $k_B T$ is much lower than the energy difference between two successive vibrational levels, so before any excitation the molecule is in the $v = 0$ vibrational level of the S_0 electronic state. In order to be absorbed, an incident photon must carry a minimum energy corresponding the $(S_0, v = 0) \rightarrow (S_1, v = 0)$ transition. A photon carrying a higher energy can also be absorbed through a transition towards the $v = 1, 2, 3\ldots$ vibrational levels of S_1. This explains the asymmetric shape of the absorption spectrum. There is a minimal energy required for a photon to be absorbed by the molecule, hence the steep slope of the low-energy (i.e. high wavelength) part of the absorption line. In contrast, higher energy photons can also absorbed, hence the smoother slope on the high-energy part of the absorption line.

Once a photon has been absorbed, the molecule is in an arbitrary vibrational level of the excited electronic state. From this level, there are two main relaxation paths: a non-radiative path through vibrational relaxation, and a radiative path through fluorescence emission. These two relaxation channels exhibit very different characteristic decay times: relaxation through vibrational states is way more faster (10^{-14} – 10^{-11} s) than fluorescence emission (10^{-9} – 10^{-7} s). As a consequence, vibrational relaxation will occur much more quickly and the molecule will relax down the vibrational ladder until it reaches

the $(S_1, v = 0)$ energy level. As vibrational relaxation can no longer occur in this state, radiative relaxation can finally take place and fluorescence emission occurs – towards any vibrational level of the fundamental electronic state. This means that fluorescence emission occurs in appreciable yield only from the lowest excited state of the same multiplicity – a principle known as Kasha's rule. Moreover, the shape of the fluorescence emission line can be understood by remarking that the maximum energy of the fluorescent photon cannot exceed the energy of the $(S_1, v = 0) \rightarrow (S_0, v = 0)$ transition, but can be lower if the relaxation occurs towards a higher vibrational level – hence the long wavelength tail of the emission peak.

Finally, let us discuss the Stokes shift. According to a rule known as the Franck-Condon principle, transitions between electronic energy states involving a change in the vibrational quantum number v (such as those sketched in Fig. 4.2) are more likely to occur than those with $\Delta v = 0$. As a consequence, multiple steps of vibrational relaxation will be observed, inducing an energy loss between the energies of the exciting photon and the fluorescence photon: the Stokes shift.

4.3 Emission and absorption of photons

The absorption rate W (units: s^{-1}) is defined as:

$$W = \sigma_{abs} \cdot \phi \tag{4.1}$$

where σ_{abs} is the absorption cross-section (units: cm^2) and ϕ is the photon flux (units: photons.cm^{-2}.s^{-1}). The emission rate k_r (units: s^{-1}) writes:

$$k_r = \frac{1}{\tau_r} \tag{4.2}$$

τ_r is the radiative decay rate, or spontaneous decay rate (units: s). If we note n the excited state population, then:

$$n(t) = n_0 \exp\left(-\frac{t}{\tau_r}\right) \tag{4.3}$$

It is possible to experimentally measure the value of τ by detecting with a high temporal resolution the number of emitted photons as a function of time (this is called *time-resolved fluorescence spectroscopy*). Typically, for organic fluorescent molecules one gets $\tau_r \sim 1$ ns.

Quantum yield

When a photon is absorbed, a fluorescence photon is not necessarily emitted. The molecule can relax to a lower energy level through non-radiative processes, that is, processes that do not involved the emission of light. Non-radiatives processes include internal conversion or vibrational relaxation (see Fig. 4.2). Thus, all molecules do not behave as efficient fluorescent emitters. The fluorescence efficiency is quantified through the quantum yield η (or quantum efficiency), defined as follows:

$$\eta = \frac{k_r}{k_r + k_{nr}} \tag{4.4}$$

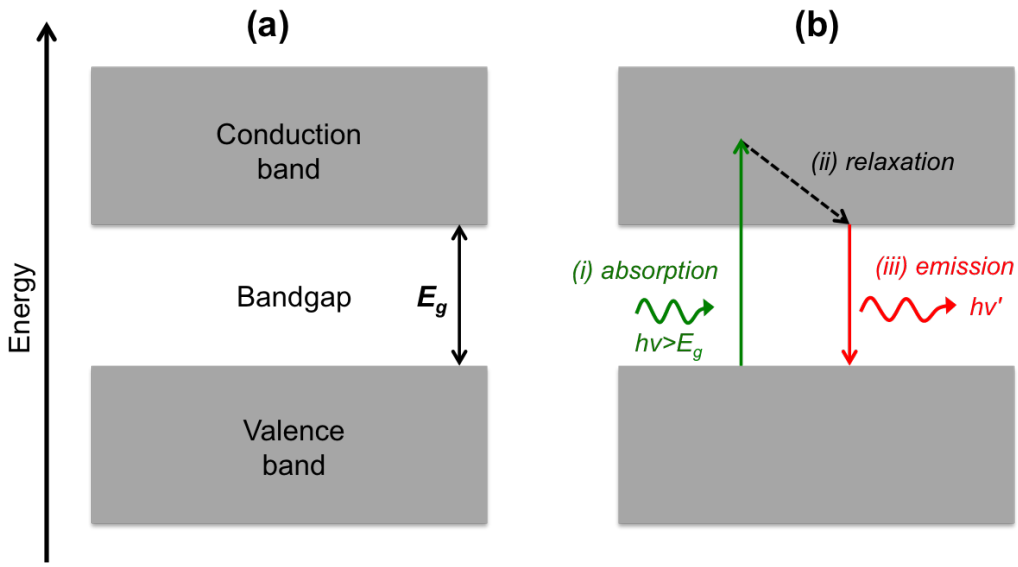

Figure 4.4: *Photoluminescence in a semiconductor. (a) Simplified sketch of the energy bands of a semiconductor. (b) Photoluminescence process: a photon with energy higher than the gap promotes an electron to the conduction band. The electron relaxes down to the bottom of the conduction band via non-radiative processes, and then recombines inside the valence band, emitting a luminescence photon whose energy is equal to the bandgap energy.*

where k_r is the radiative emission rate and k_{nr} account for all non-radiative relaxations. A perfect emitter would exhibit $\eta = 1$. A very good emitter like the Rhodamine6G molecule (diluted in ethanol) exhibits $\eta = 0.95$, but most emitters have lower quantum yields. From the quantum yield it is also convenient to define the fluorescence cross-section $\sigma_{fluo} = \sigma_{abs} \cdot \eta$.

4.4 Photoluminescence in solids

In spite of their different nature, photoluminescence in solid-state emitters can be described in a very similar way than molecular fluorescence. Two kinds of solid-state emitters can be distinguished. First, photoluminescence emission can be observed in bulk solids – generally semiconductors (such as GaAs, ZnO, GaN...). Second, it can be observed in specially engineered nanostructures, such as quantum wells or quantum dots, whose tiny dimensions allows the quantum confinement of electrons. Quantum dots are sometimes called "artificial atoms", because they exhibit discrete energy levels, similarly to atoms or molecules.

4.4.1 Photoluminescence in bulk semiconductors

In contrast with the discrete energy levels of molecules, electronic energy levels in solids are organized into energy bands, as sketched in Fig. 4.4. Those bands are filled by electrons until the valence band, which is playing a role similar to the HOMO in a molecule, while the role of the LUMO is played by the conduction band. Those two bands are separated by an energy difference E_g called the bandgap, or gap. Please note that is a simplified

version of the energy band model, overlooking the role played by the different directions inside the crystalline structure of the solid[1]. However, this simple model is sufficient to understand how photoluminescence happens in semiconductors.

An incoming photon, with a higher energy than the gap energy, $E_{ph} > E_g$, can be absorbed inside the semiconductor. This promotes an electron from the valence to the conduction band. The promoted electron leaves a vacation in the valence band, called a hole. The electron-hole pair forms a quasi-particle called an exciton. Similarly to what happens in molecules, the electron will first relax down to the bottom of the conduction band, losing any exceeding energy into vibrations of the crystalline lattice. These lattice vibrations are generally modelled as quasi-particles called phonons, which are quanta of vibrational energy. Finally, the electron-hole pair will recombine (i.e. the electron will relax to the valence band), either radiatively (emitting a luminescence photon) or non-radiatively.

4.4.2 Solid-state quantum emitters

A solid-state quantum emitter is very similar to a fluorescent molecule, in the sense it exhibits discrete energy levels. Consequently, most of the notions we introduced for fluorescent molecules can be directly translated to solid-state quantum emitters: absorption cross-section, radiative and non-radiative rates, quantum yield and saturation intensity can all be used.

Discrete levels can appear in the band structure of a solid due to defects in the crystalline structure. *Color centers* (or F-centers) are optically active defects appearing in insulators. Color centers exist naturally in solids, but can also be engineered, either by the controlled growth of synthetic crystals or by irradiation with particles such as electrons. The nitrogen vacancy (NV) center in diamond is good example of such a defect. A NV center appears when a carbon atom in the diamond lattice is substituted by a nitrogen atom, leaving a vacancy in an adjacent lattice site. NV centers exhibit a characteristic sharp emission line at a wavelength $\lambda = 637$ nm under green excitation. They are subject of an intense attention due to their potential applications in quantum technologies.

Discrete energy levels can also appear in nanostructures in which the electrons and holes are confined in one or more direction. This effect is known as quantum confinement and can be understood using the "electron-in-a-box" model. If the size of the box is small enough (smaller than the Bohr radius of the confined exciton), then quantum mechanics imposes the existence of discrete energy levels, whose positions are controlled by the size of the box. This is the main advantage brought by quantum-confined solid-state emitters: their emission wavelength can be tuned by changing their characteristic size. Colloidal quantum dots are probably the most ubiquitous solid-state quantum emitters. They consist of a few nanometers semiconductor nanoparticles (generally spherical), allowing for three-dimensional quantum confinement. Most common materials include CdSe and PbS, yielding emission wavelengths covering a broad region of the spectrum, from near ultraviolet to infrared with high quantum yields.

[1]For a more detailed presentation, see M. Fox, *Optical properties of solids* (2nd edition), Oxford University Press, 2010.